

## Challenge: Strings it

**Category:** General Skills

**Description:** The challenge provided a downloadable file and asked to find the flag without running/executing it.

**Approach:** After downloading the file, I confirmed its type using `file`, which showed it was an ELF 64-bit executable (a compiled Linux binary), not stripped of its debug symbols. Since running the binary directly wasn't allowed, I used the `strings` command to extract all readable, printable text from it without executing it. My initial filter using `grep -i pico` returned no results, so I broadened the search using `grep "{"` — searching for the curly brace character used in flag formatting generally, rather than assuming a specific prefix — which successfully located the flag.

**Difficulty Faced:** My first `wget` attempt downloaded an HTML page instead of the actual binary. I resolved this by first opening the challenge file link in a browser on Windows to confirm the correct file, then downloading it directly through Kali's browser instead, ensuring I had the real binary locally rather than a redirect/login page. I also initially filtered with `grep -i pico`, assuming the flag would follow the standard `picoCTF{...}` format, but this platform used `academy{...}` instead — broadening my search to the `{` character resolved this.

**Tools used:** `wget`/browser download, `file`, `strings`, `grep`

**Flag:** `academy{5tRIng5_1T_8552bB3B}`

**What I learned:** The `strings` command is a genuinely powerful, low-risk way to inspect a binary without executing it. I also learned that flag formats can vary between platforms, so filtering too narrowly based on an assumption can cause you to miss something that's actually right there — a more general search pattern is often safer when you're not certain of the exact format.

**Evidence:**

```
(kali㉿kali)-[~/Downloads]
└─$ strings strings | grep -i {
academy{5tRIng5_1T_8552bB3B}
```